

Ethanol Blending in Gasoline – Japan

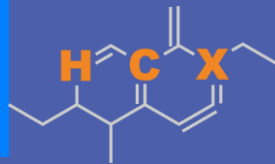
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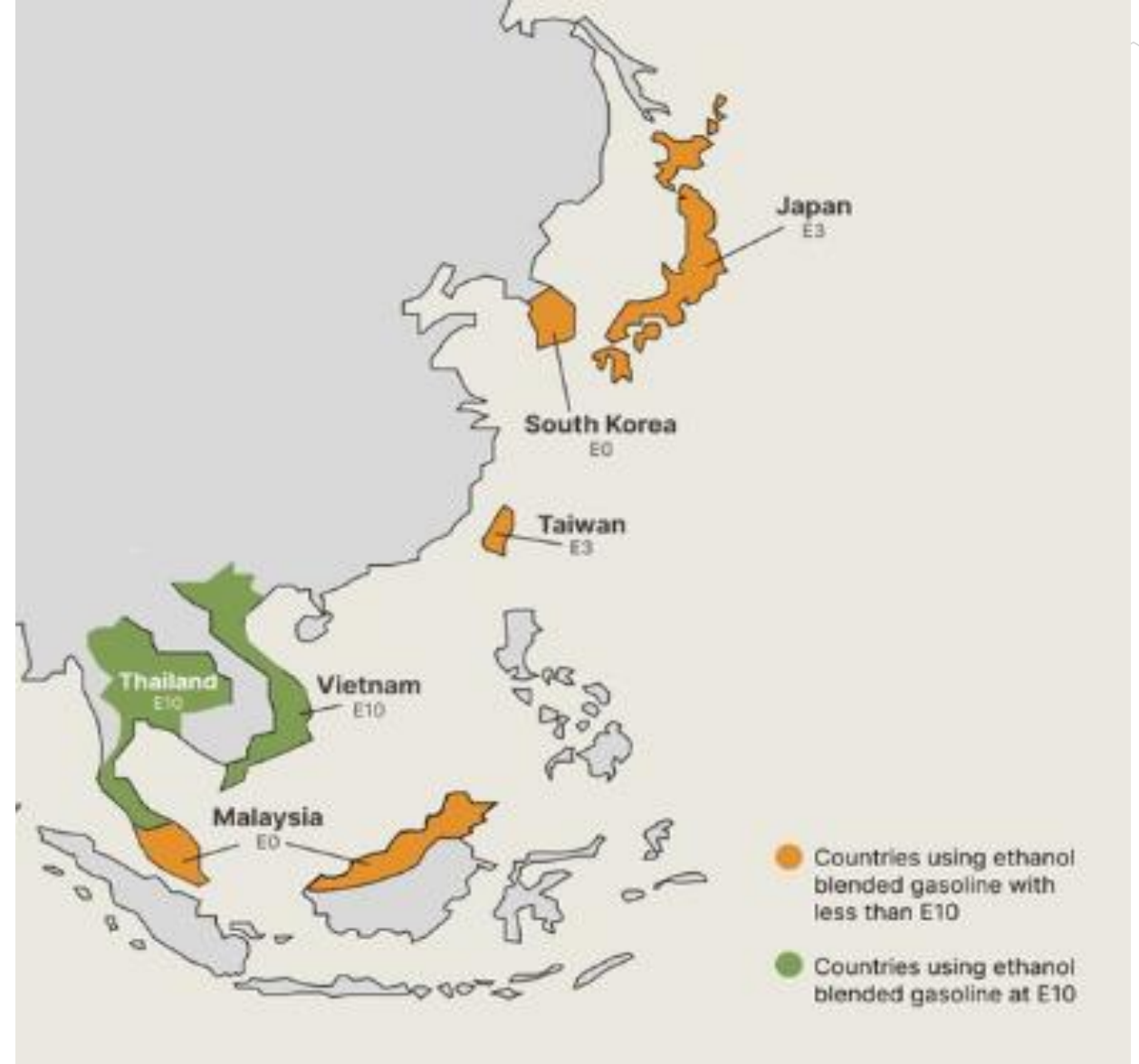


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Japan



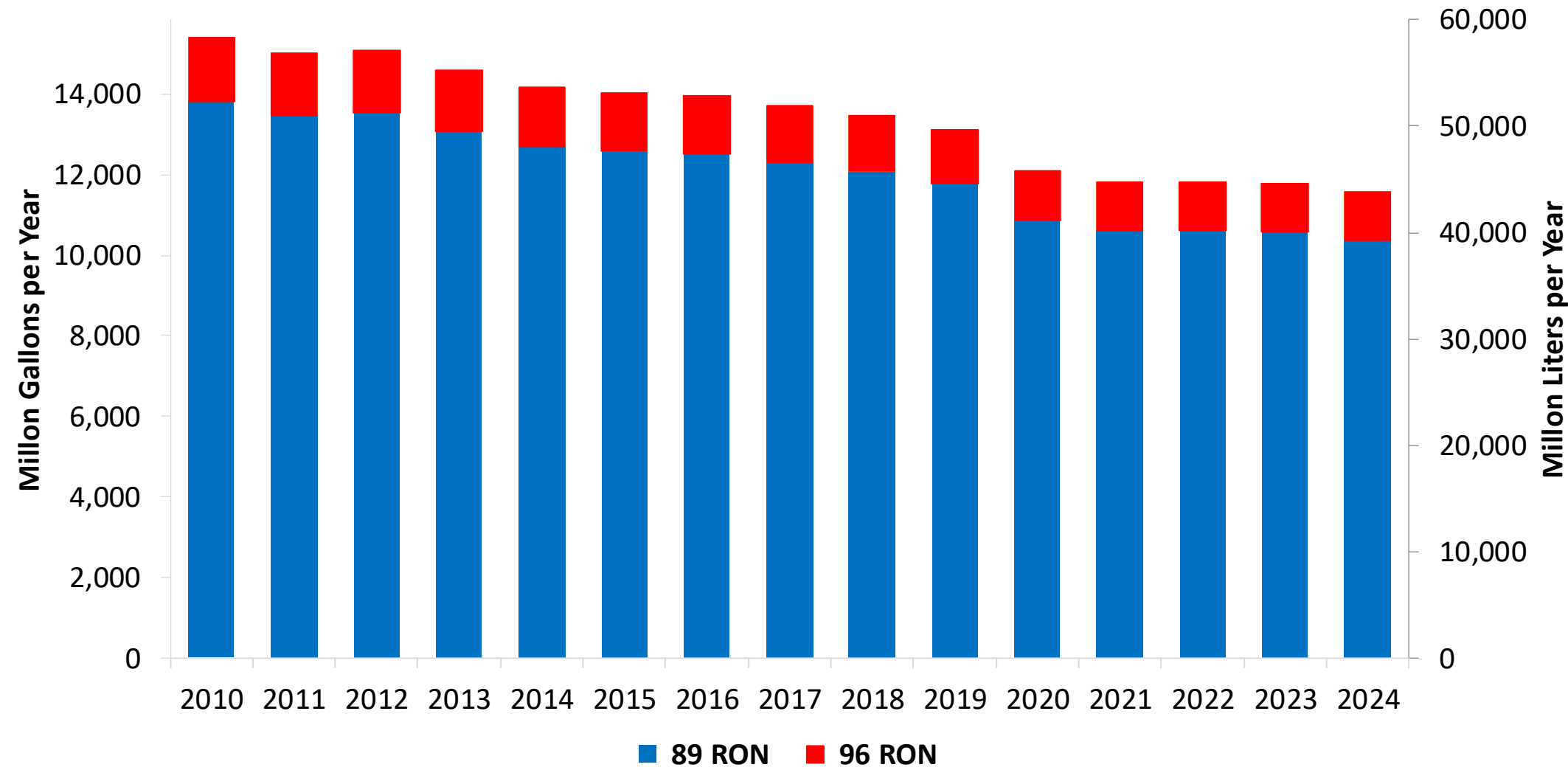
In 2024, gasoline consumption in Japan was 43,826 million liters (11,578 million gallons) and growth rate was -2.5%. Gasoline production and net imports were 42,543 and 1,283 million liters (11,239 and 339 million gallons) correspondingly. Japan complies with the JIS standards and offers two main grades: RON 89 and RON 96) with an average octane of 89.7 RON.

Japan currently allows E3. Last year's consumption was 830 million liters (219 million gallons) equivalent to 1.9%, primarily as the component ethyl tert-butyl ether (ETBE). There is no local production. Ethanol is imported as a blendstock for ETBE production and as ETBE. The Ministry of Economy, Trade and Industry (METI) aiming for E10 (10% ethanol) by 2030 and E20 (20% ethanol) by 2040. This initiative requires infrastructure improvements, as most existing gas stations are only equipped for the E3 blend.

Source: METI, USDA

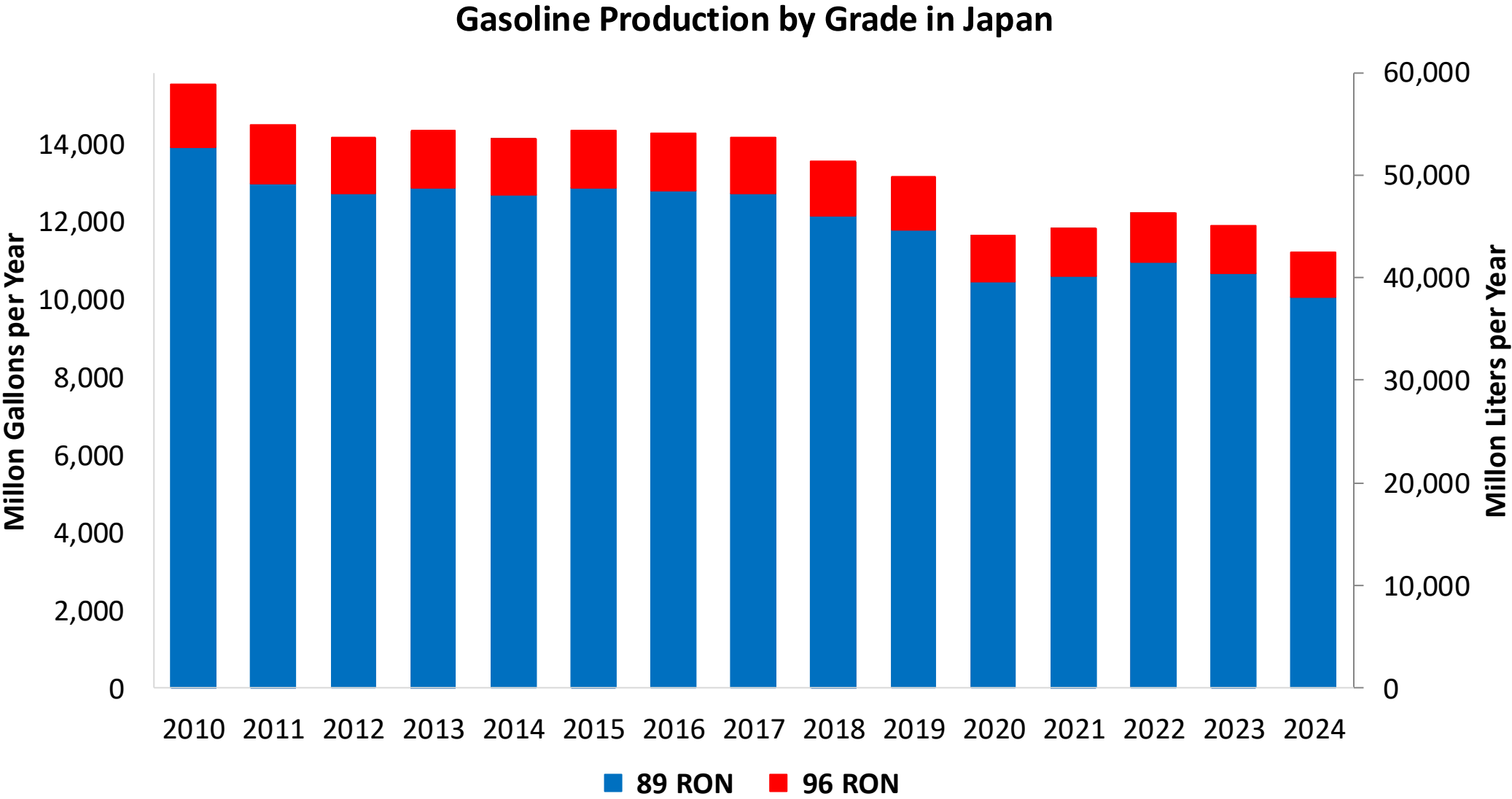
Gasoline Demand in Japan

Gasoline Demand by Grade in Japan



Source: METI

Gasoline Production in Japan



Source: METI

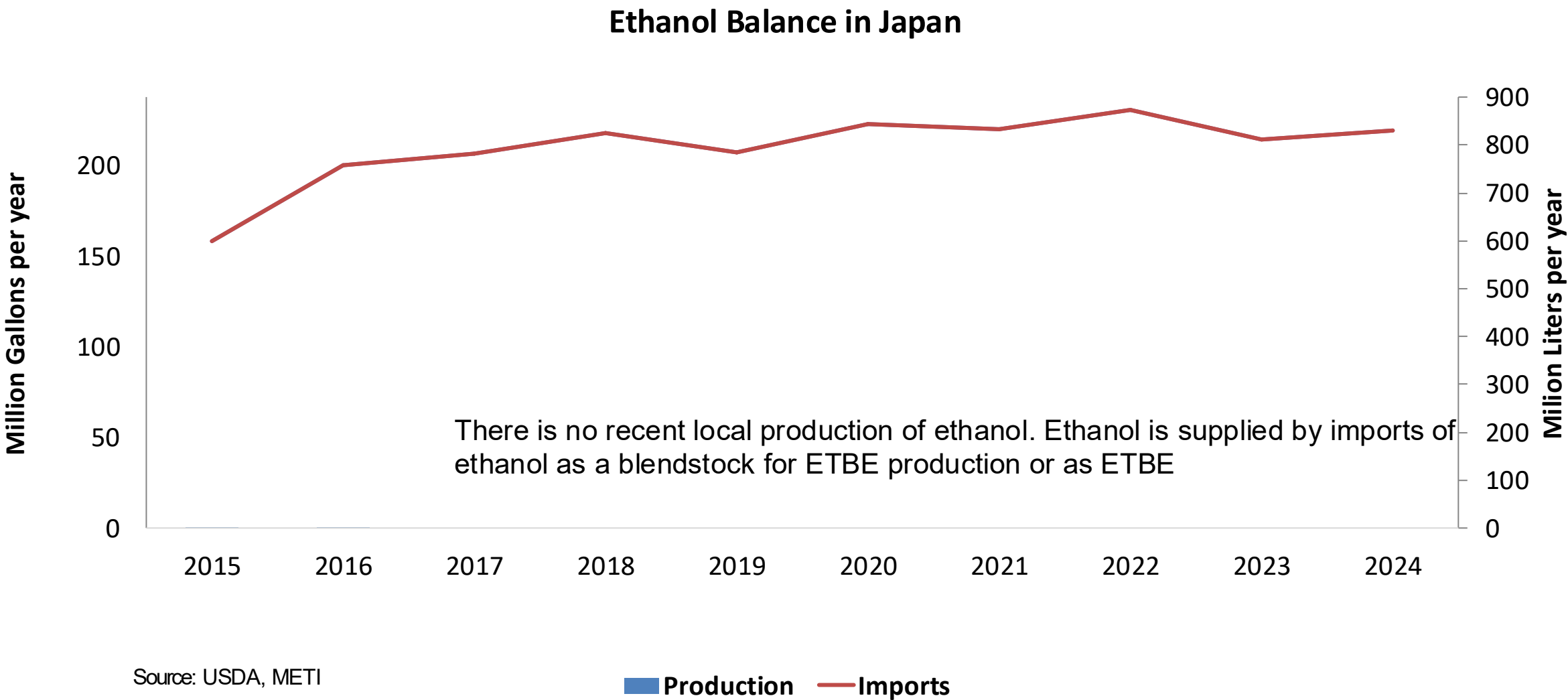
The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a methyl group (C), and an ethyl group (X) at various positions on the ring.

The chart displays the annual consumption of 89 RON and 96 RON gasoline from 2010 to 2024. The 89 RON consumption (blue) shows a significant increase starting in 2018, peaking in 2024. The 96 RON consumption (red) remains relatively stable, contributing a smaller portion of the total volume.

Year	89 RON (Million Gallons)	96 RON (Million Gallons)	Total (Million Gallons)
2010	260	30	290
2011	600	70	670
2012	660	80	740
2013	450	50	500
2014	380	40	420
2015	300	40	340
2016	200	20	220
2017	230	30	260
2018	490	60	550
2019	540	60	600
2020	740	90	830
2021	690	90	780
2022	550	70	620
2023	760	90	850
2024	1130	140	1270

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Ethanol Balance in Japan



Gasoline Quality in Japan

Name	JIS K 2202: 2023 - Law on the Quality Control of Gasoline and Other Fuels							
Implementation Date	2023							
Applicability	Japan							
Grade	Regular		Premium		E10 Regular *		E10 Premium *	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	89.0		95.0		90.0		96.0	
MON								
AKI								
Benzene (%vol)		1.0		1.0		1.0		1.0
Aromatics (% vol)		-		-		-		-
Olefins (%vol)		-		-		-		-
Lead (g/L)		-		-		-		-
Manganese (mg/L)		-		-		-		-
Sulfur (ppm)		10		10		10		10
Oxygen (%w)		1.3		1.3		3.5		3.5
Ethanol (%vol)		3.0		3.0	3	10.0	3.0	10.0
MTBE (%vol)		7.0		7.0				
PVR 37.8°C (psi)	6.4	11.3	6.4	11.3	6.4	11.3	6.4	11.3
Carbon Intensity (gCOe/MJ)	0	88.74	-	88.74	0	88.74	-	88.74
Kerosene (%vol)	-	4.0	-	4.0	-	4.0	-	4.0

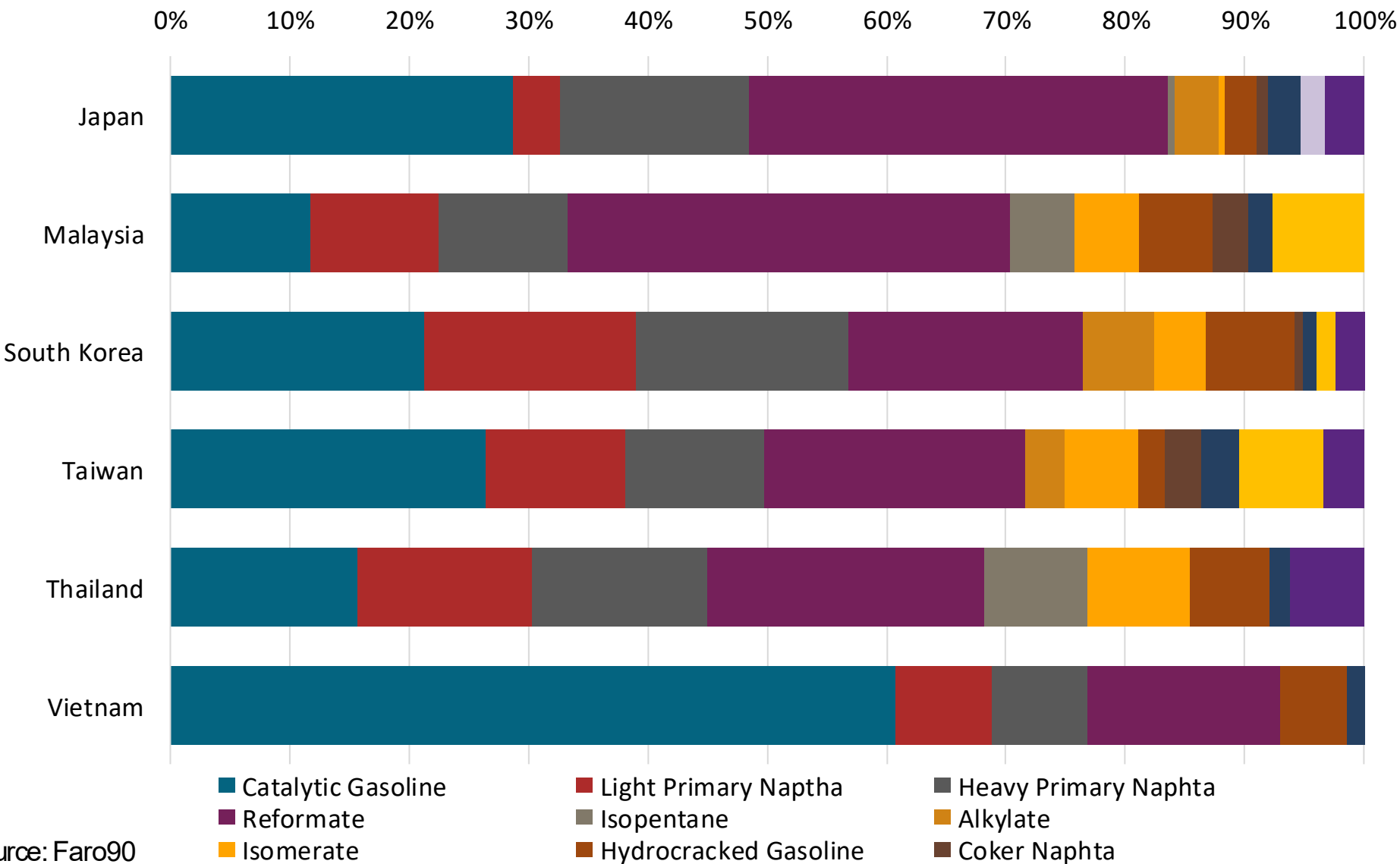
*Not Mandatory. A 55% Carbon Intensity Reduction target has been established and will be revised once LCA has been completed

Source: METI

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



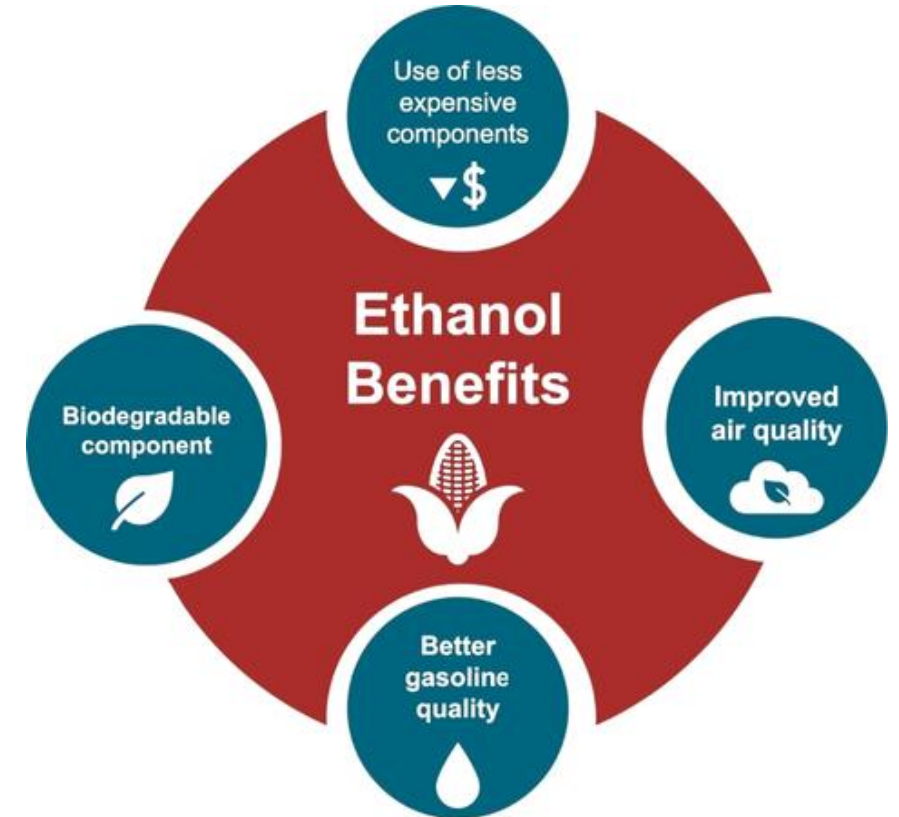
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

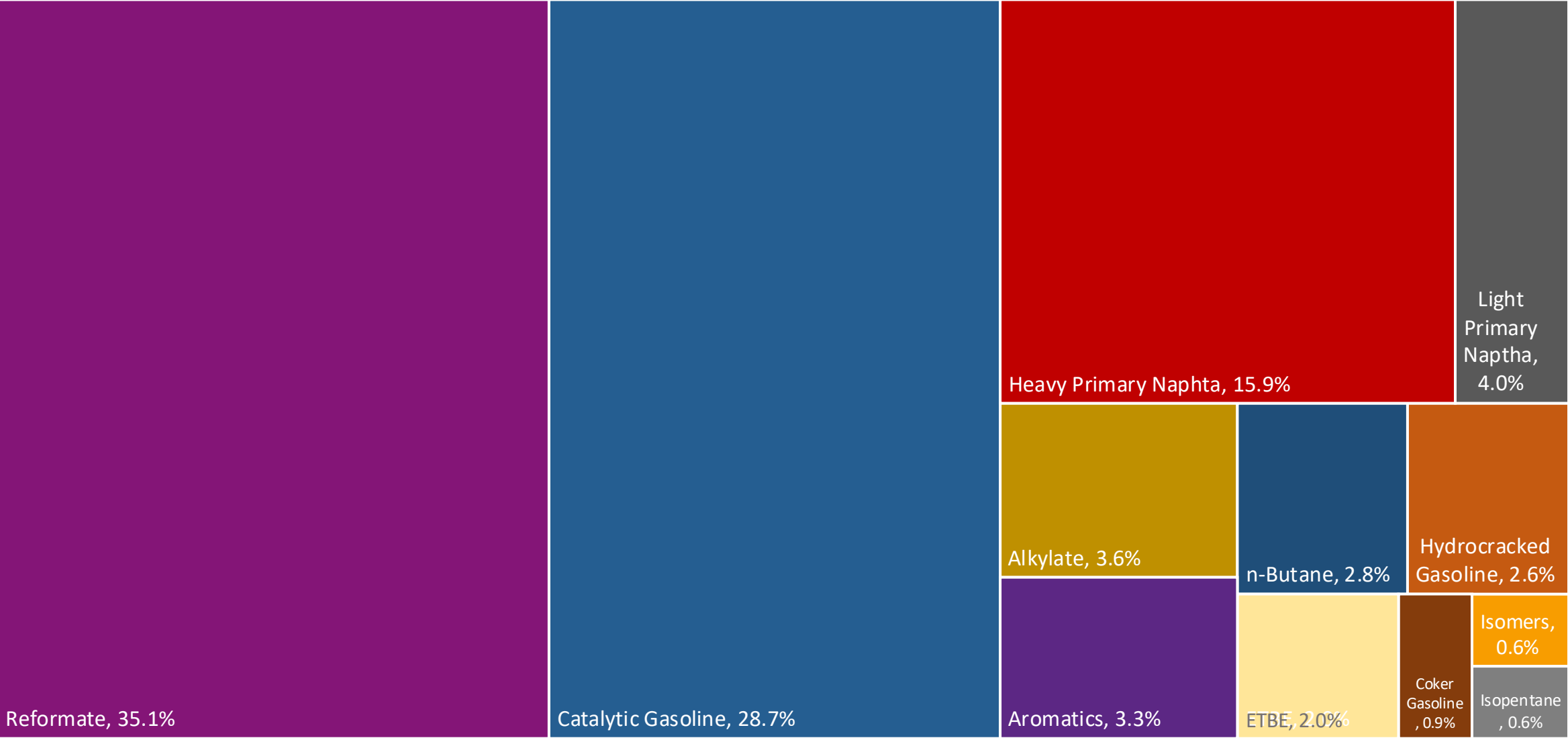
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

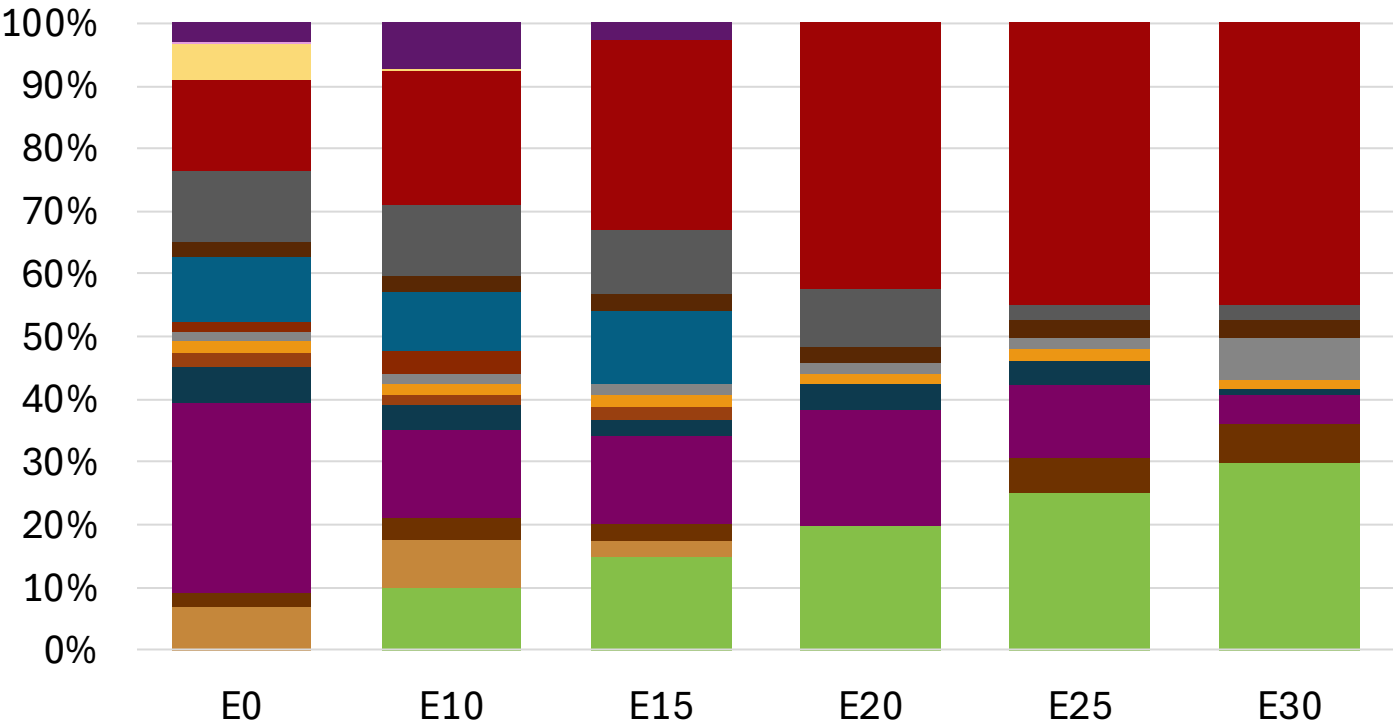


Japan Available Blending Components



Source: Faro90

Japan – Gasoline 1 – Constant Octane

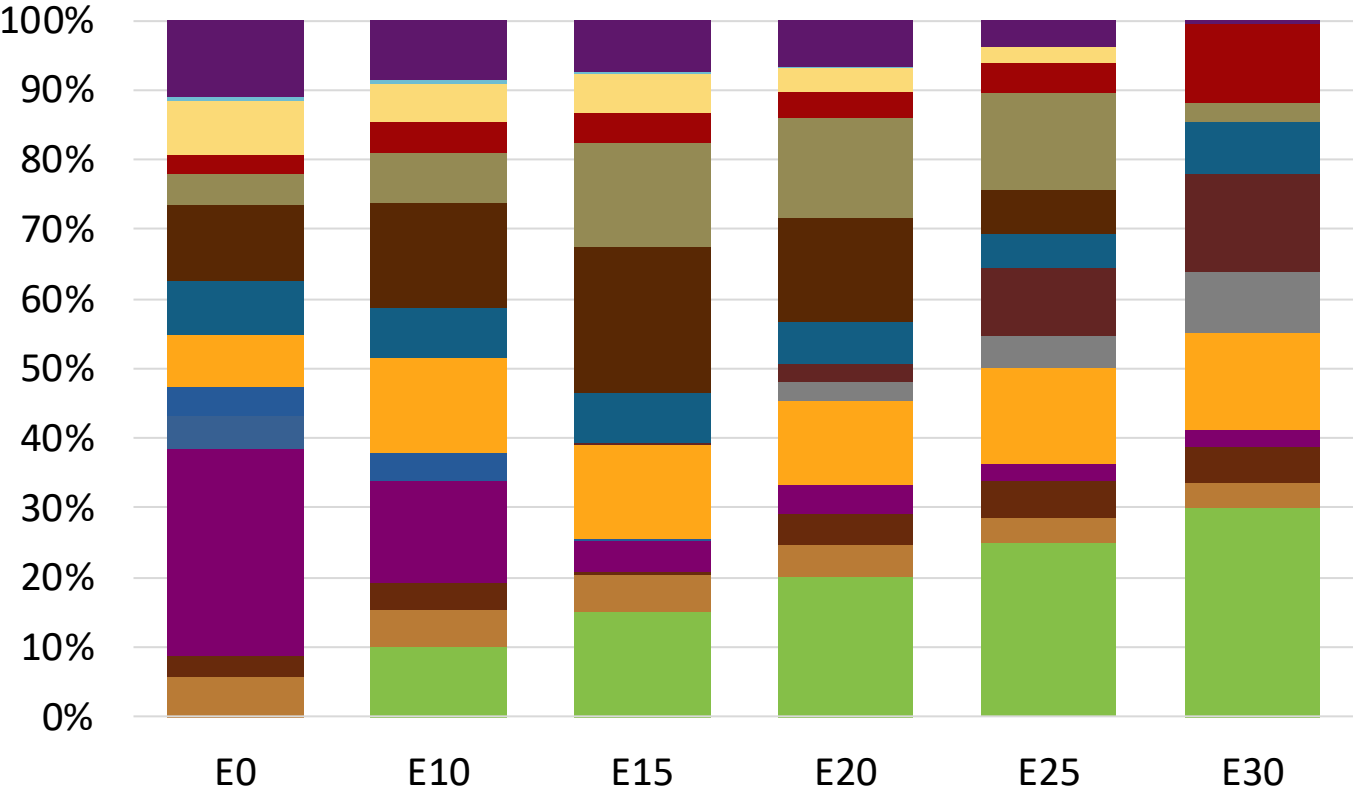


Average CIF 2024 Prices

Octane (RON)	89.0	89.0	89.0	89.0	90.0	90.7
Price (US\$/gal)	\$ 2.223	\$ 1.969	\$ 1.826	\$ 1.678	\$ 1.611	\$ 1.530

Source: Faro90

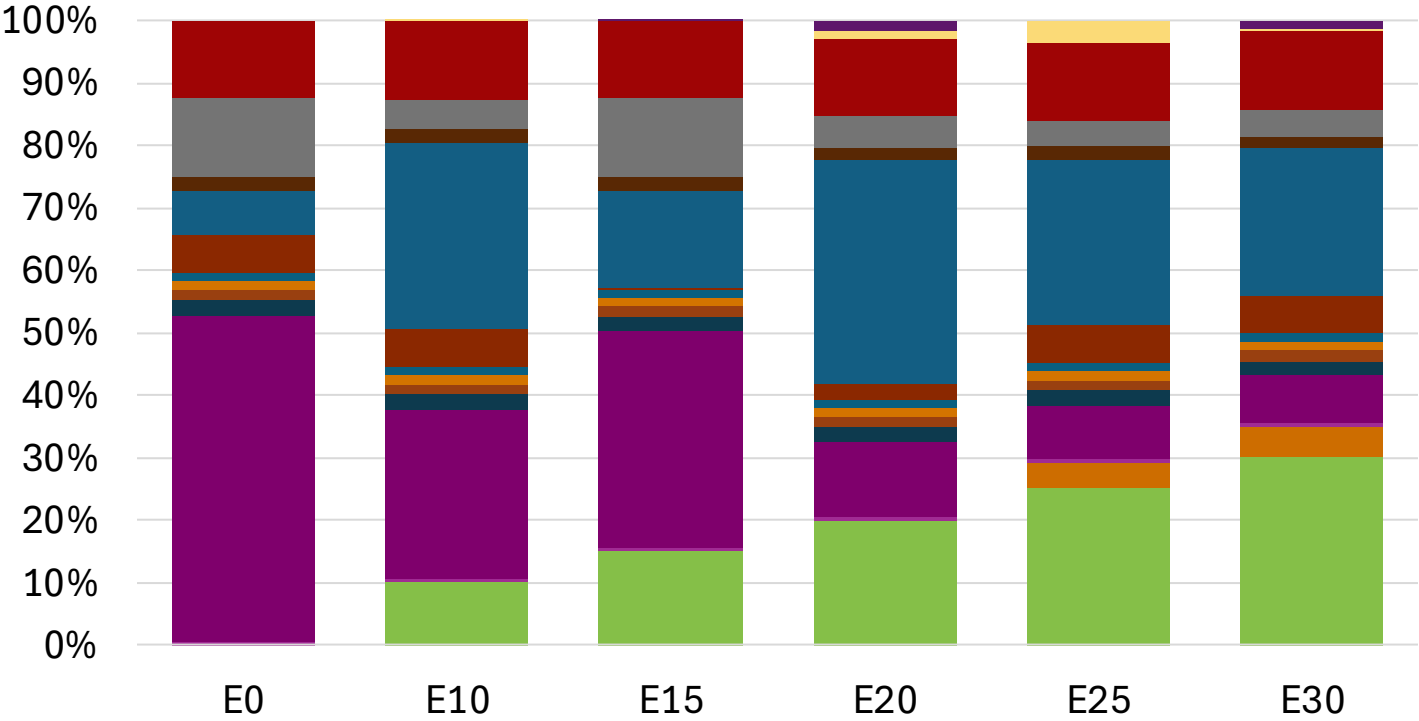
Japan - Gasoline 2 - Constant Octane



Average CIF 2024
Prices

Octane (RON)	96.0					
Price (US\$/gal)	\$ 2.421	\$ 2.344	\$ 2.265	\$ 2.180	\$ 2.090	\$ 1.984

Japan – Gasoline 1 – Increased Octane

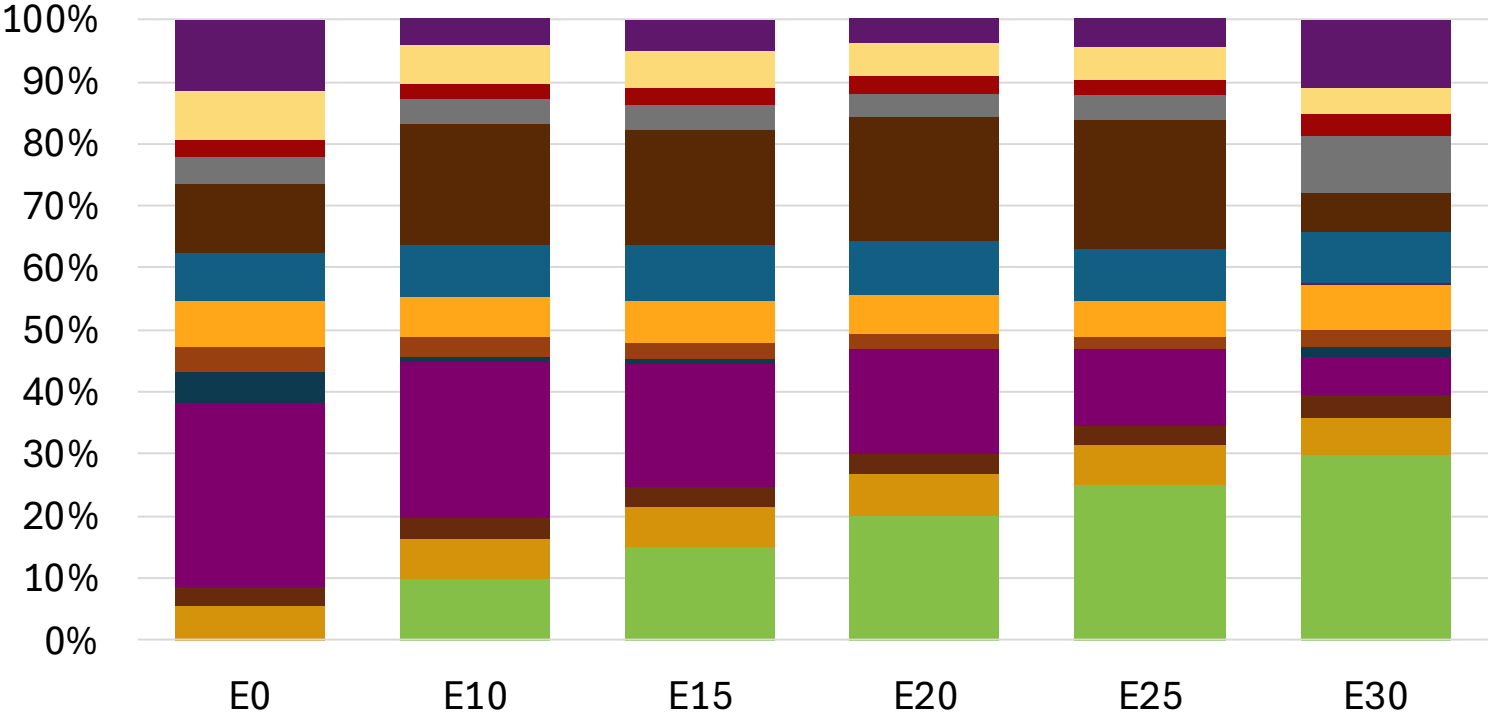


Average CIF 2024
Prices

Octane (RON)	89.2	93.3	95.2	97.1	98.8	100.0
Price (US\$/gal)	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223

Source: Faro90

Japan – Gasoline 2 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	96.0	97.6	99.1	100.5	102.0	103.2
Price (US\$/gal)	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421	\$ 2.421

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

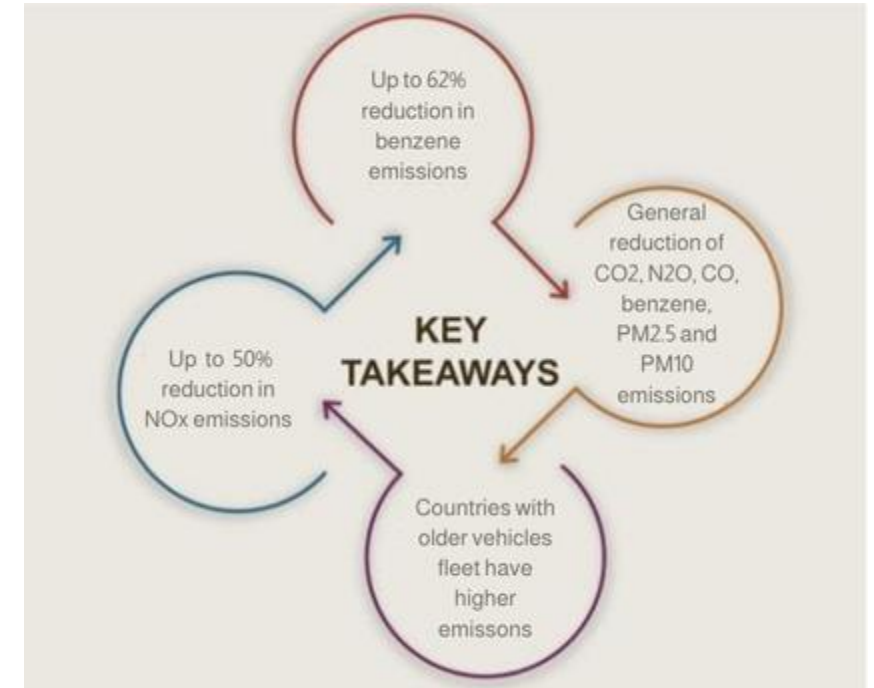
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

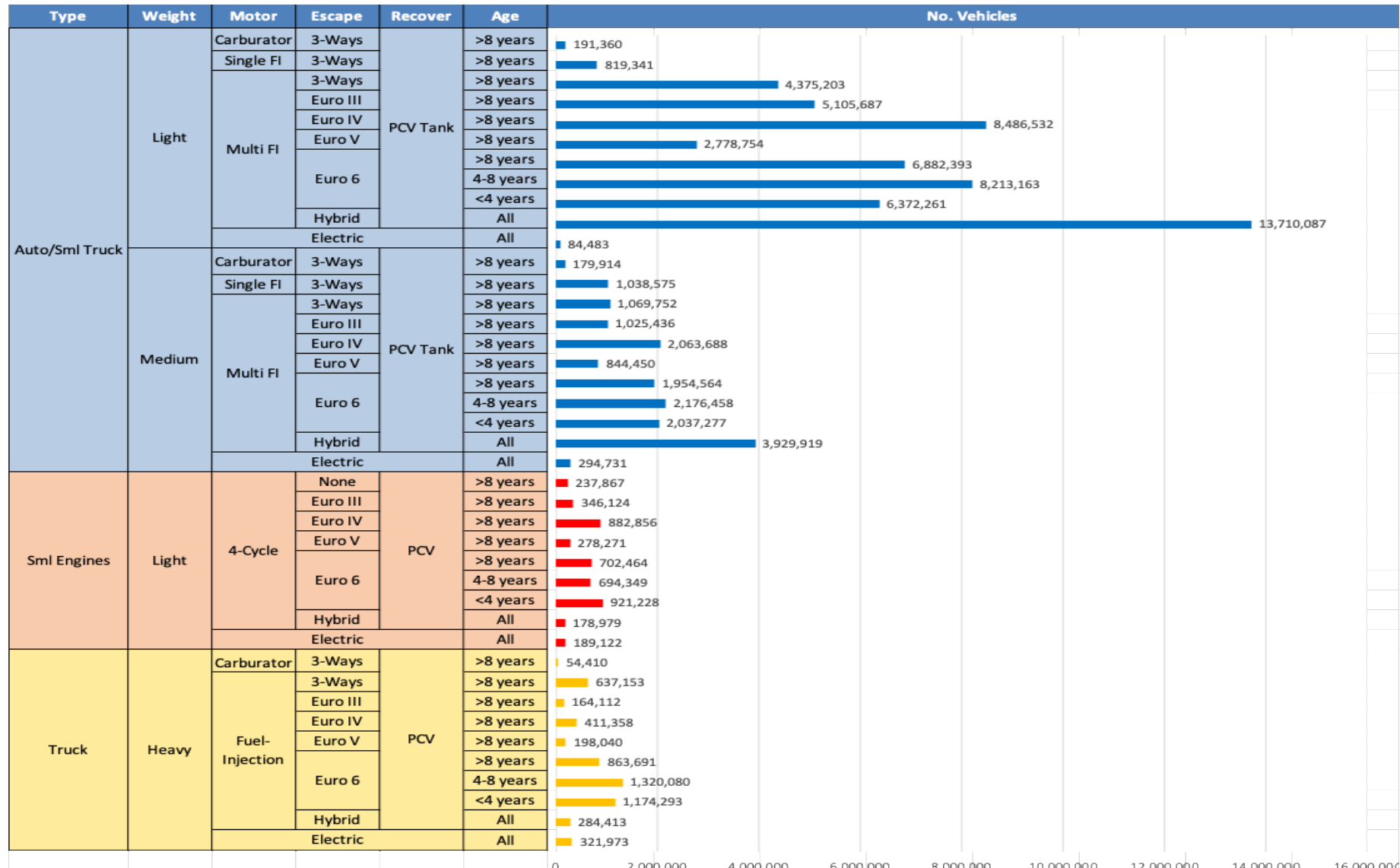
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Japan

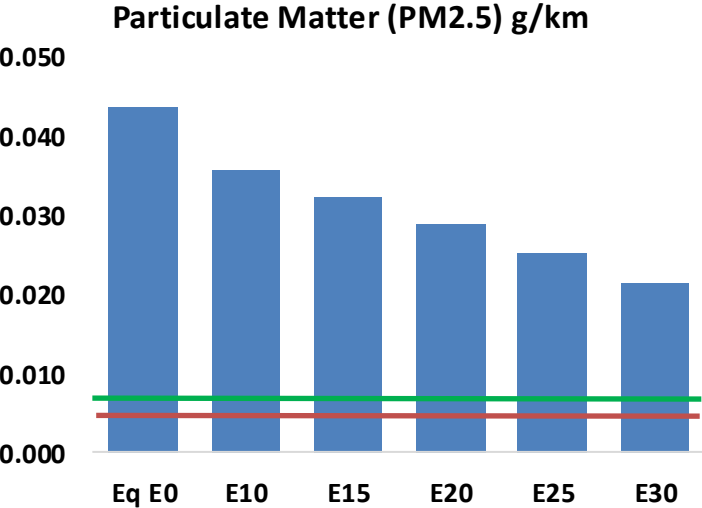
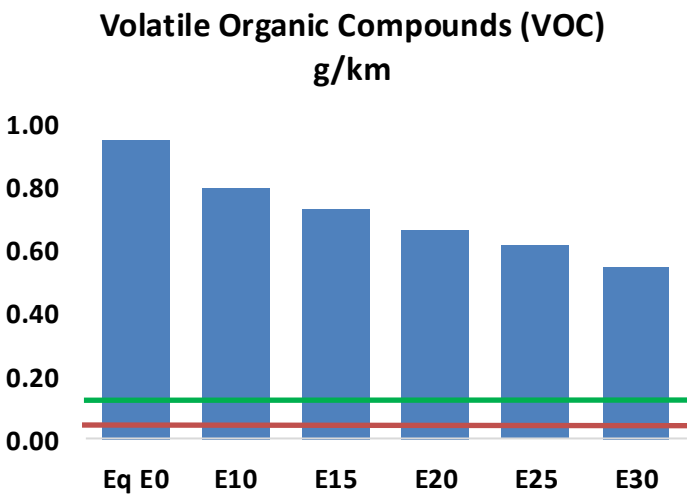
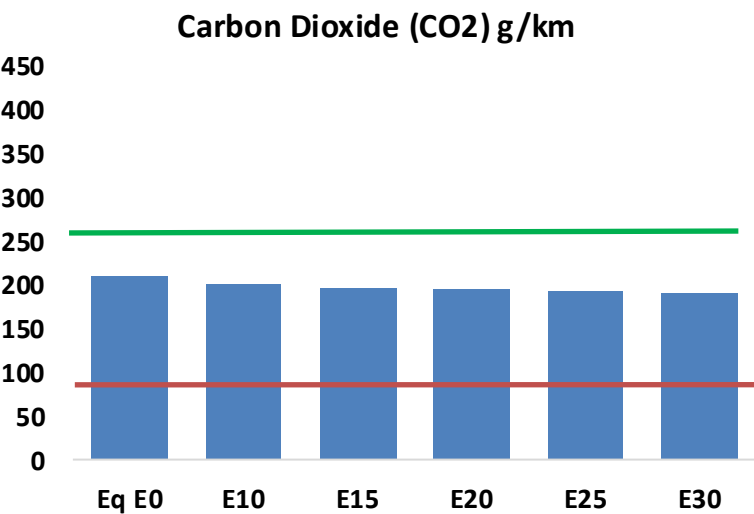
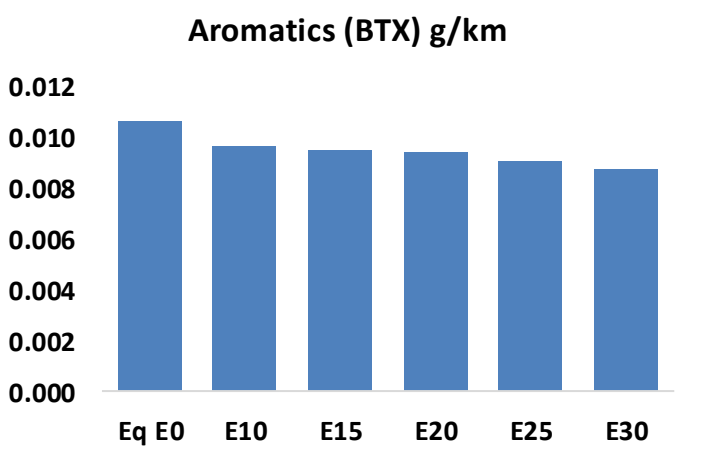
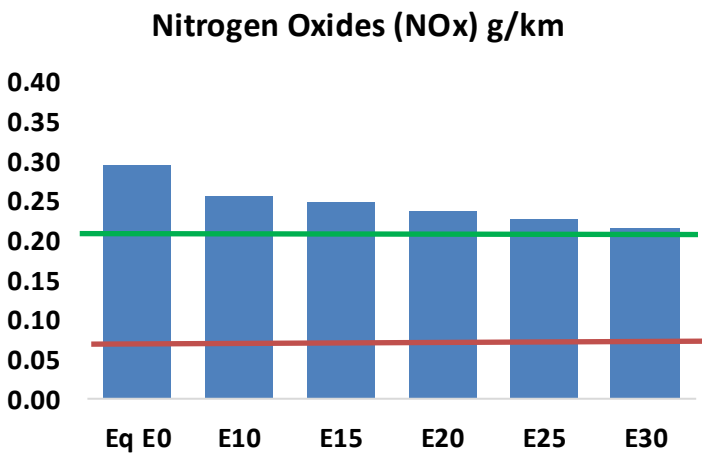
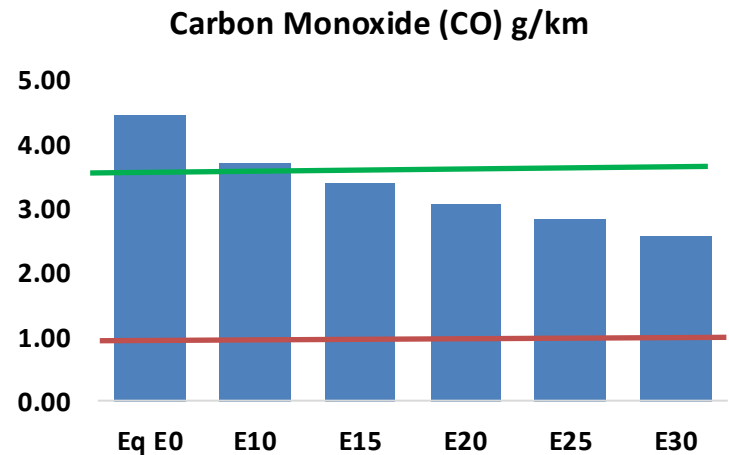
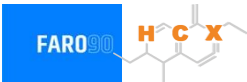


Vehicle Fleet: 83,494,811
Gasoline Vehicle Fleet: 76,775,958

Source: AIRIA

Japan – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Japan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	7,996,709	7,320,077	6,643,444	6,076,898	5,501,211	5,056,863	4,584,871	-17%	-31%
CO2	375,859,078	366,194,131	357,066,124	349,887,216	346,334,811	344,557,634	338,206,446	-5%	-8%
VOC	1,702,687	1,563,867	1,425,046	1,309,743	1,192,997	1,103,012	983,689	-16%	-30%
NOx	529,045	488,485	460,752	446,675	423,941	406,147	385,180	-13%	-20%
SOx	4,260	3,938	3,616	3,343	3,082	2,799	2,501	-15%	-28%
NH3	146,209	144,761	143,314	143,994	145,614	146,753	147,700	-2%	0%
N2O	7,310	7,270	7,241	7,276	7,426	7,509	7,594	-1%	2%
CH4	345,533	345,533	345,533	350,716	358,317	364,882	371,102	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	6,553	6,230	5,906	5,675	5,460	5,299	5,067	-10%	-17%
Acetaldehyde	23,643	27,135	39,815	60,634	82,607	94,394	111,537	68%	249%
Formaldehyde	91,090	88,560	103,236	121,221	126,997	140,493	152,942	13%	39%
Benzene	19,076	18,332	17,370	17,095	16,953	16,214	15,690	-9%	-11%
PM2.5	22,257	20,234	18,211	16,430	14,688	12,820	10,866	-18%	-34%
PM10	55,998	50,908	45,817	41,337	36,954	32,255	27,337	-18%	-34%

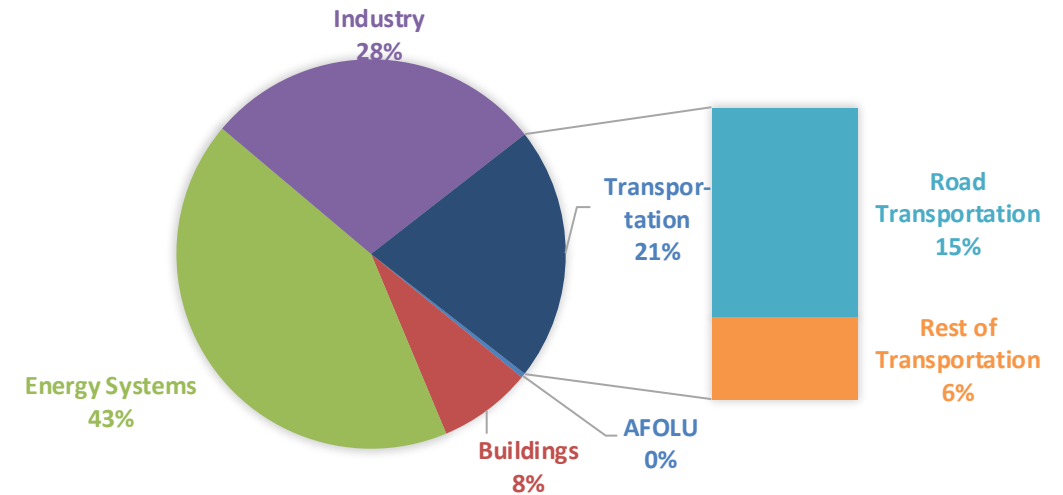
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

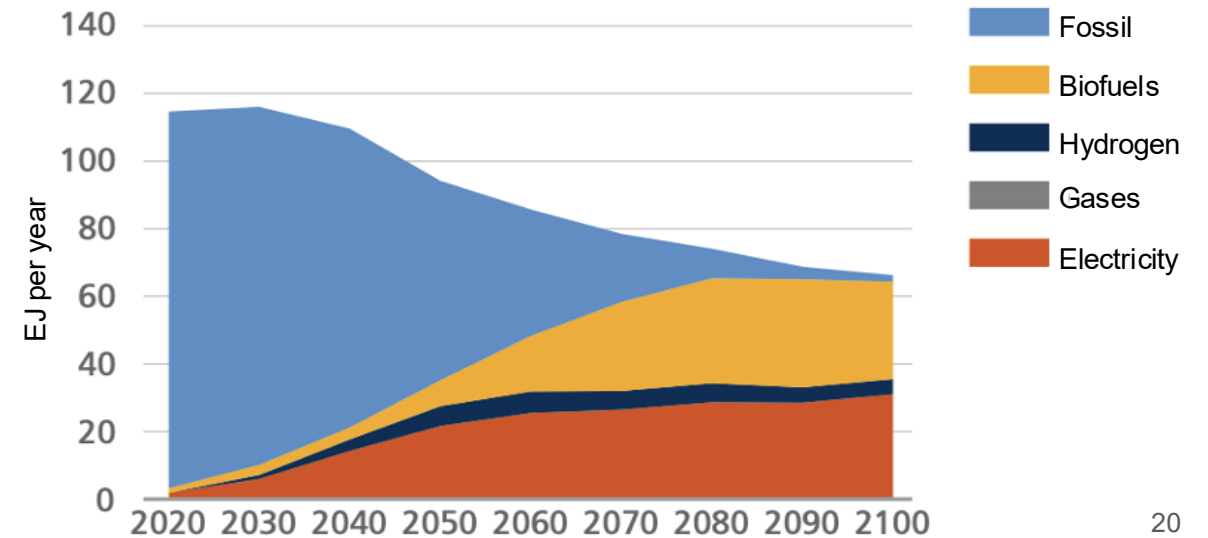
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

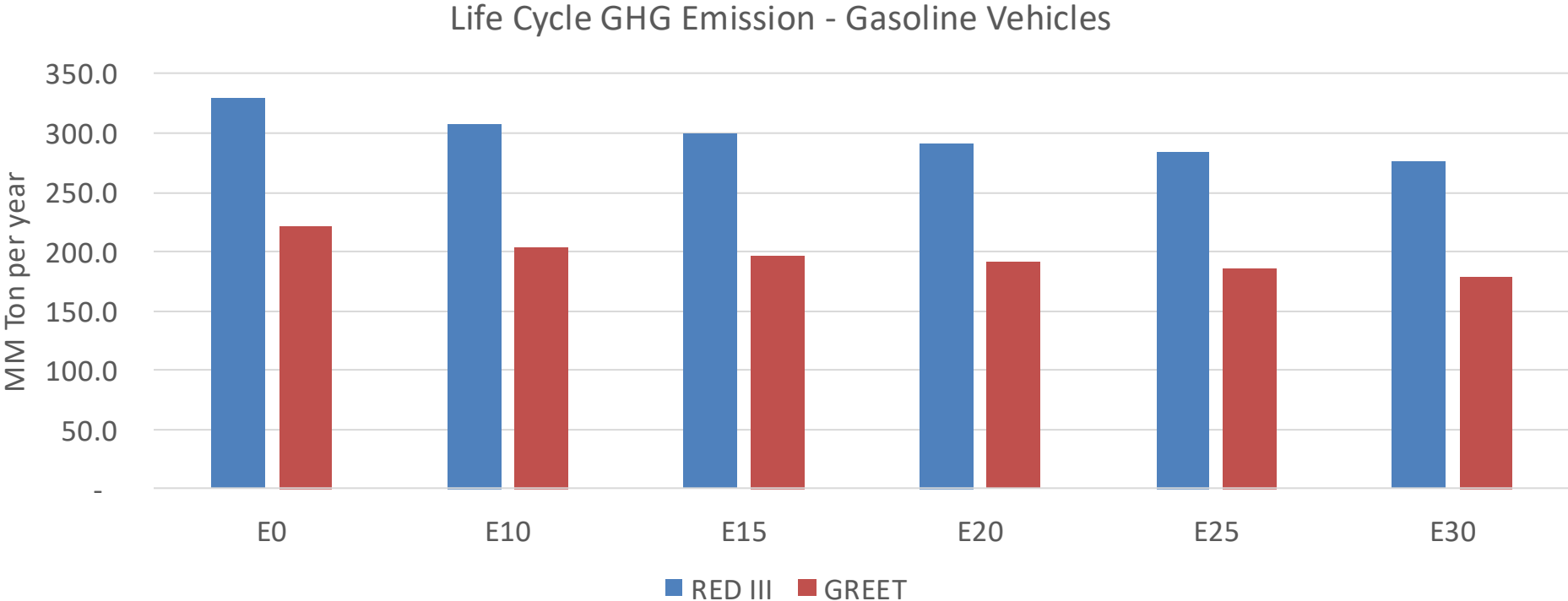
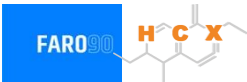
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Japan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/year)	1,102.2
Target @ 2035 (MMT/year)	570.4
Delta	531.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.8%	9.3%	11.6%	13.6%	16.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.8%	6.7%	8.0%
RED II/III	0.0%	4.2%	5.7%	7.2%	8.4%	10.1%

Source: Greet, RED II/III, climateactiontracker.org, F90